VIVEK K.

MSc Robotics, Uni. of Sheffield,| 2+YoE | Control System Design | Embedded C | Embedded & Control Systems Engineer
UK | +44 7407 853032 | Gmail | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Certified in Industrial Automation and Level 7 qualified in Robotics, with **2+ years' experience** applying Siemens S7 PLC programming, SCADA integration and electrical fault diagnosis to restore **2** crash-prone control systems to stable operation. **Recognised for taking ownership of complex PLC recovery and troubleshooting initiatives,** restoring two critical systems to stable operation.

RELEVANT SKILLS

- **Functional Competencies:** Control System Design | Closed-Loop Control | PID Control & Tuning | Industrial Automation | PLC Programming | SCADA & HMI Integration | Commissioning | Fault Diagnosis | Electrical Schematic Interpretation | Hardware Integration & Testing | Sensor & Actuator Integration

- **Technical Tools:** Embedded C | STM32 | STM32CubeIDE | Siemens S7 PLC | SCADA | HMI | Arduino Uno/Mega | MATLAB/Simulink | ROS | Python | Git | Fusion 360 | PCB Design

WORK EXPERIENCE

Engineer, Indo-German Tool Room Dec 2022 – Aug 2023

- Took ownership of 2 neglected Siemens S7 PLC systems by obtaining approval to investigate faults beyond the trainee remit, resulting in a structured recovery effort for both systems.
- Diagnosed and repaired recurring PLC failures by tracing wiring faults and replacing degraded components, restoring 2 crash-prone systems to stable operation.
- Recalibrated repaired control systems by applying industrial operating standards and validation checks, enabling reliable use in practical training for 50+ engineering students.
- Created 2 technical lab manuals by documenting PLC and embedded-system procedures, improving consistency in commissioning, troubleshooting, and training activities.
- Supported commissioning and maintenance activities by performing Siemens S7 programming, wiring verification, and functional testing, helping ensure consistent post-repair system performance.

Automation Team Member, TechMission Jan 2022 – Apr 2022

- Resolved a conveyor sensor issue by tracing electrical schematics and assisting with sensor replacement and alignment, restoring normal process runtime at the client site.
- Developed 3 of 8 SCADA monitoring screens by configuring interfaces and integrating them with the control system, delivering operator visibility for 2 external client companies.
- Integrated SCADA components with PLC-based systems by performing interface configuration and control checks, improving reliable communication between monitoring and control functions.
- Completed panel wiring and schematic verification by checking connections against electrical drawings, ensuring accurate hardware implementation for automation projects.
- Contributed to PLC and SCADA development by assisting with programming, wiring, and testing activities, strengthening end-to-end industrial automation operations.

PROJECTS

PID Motor Speed Controller, Personal Project 2026 – Present
Technologies: STM32, Embedded C, STM32CubeIDE, Quadrature Encoder, PWM, PID Control

- Designed a closed-loop motor speed control system by implementing a PID algorithm from scratch in Embedded C on an STM32 microcontroller, enabling real-time RPM regulation using encoder feedback.
- Developed the control firmware by configuring timers, interrupts, PWM generation, and encoder interfaces in STM32CubeIDE, creating a responsive and deterministic motor-control loop.

Automated Agitation Platform, University of Sheffield Mar 2024 – Feb 2025
Technologies: Arduino Mega, Embedded C, Mechanical Design

- Designed and built an automated agitation platform by integrating an Arduino Mega with a custom 8-sided rotating mechanism, replacing a manual assembly process and reducing cycle time from 2–3 minutes to 1–1.5 minutes.
- Developed the embedded control logic by programming rotation sequences, timing control, and actuator coordination in Embedded C, ensuring repeatable part positioning during operation.

Autonomous Exploration & SLAM, University of Sheffield Feb 2024 – Apr 2024
Technologies: ROS, SLAM, Occupancy Grid Mapping, Path Planning

- Developed an autonomous exploration system by contributing to ROS-based navigation, mapping, and path-planning modules, enabling coordinated exploration of an unknown environment.
- Implemented a trap-recovery algorithm by using a turn-counter to detect repetitive circling behaviour and trigger corrective manoeuvres, eliminating a major failure mode that previously stalled exploration.

Smart Helmet for Rider Safety, Shri Vaishnav Institute of Technology and Science Aug 2021 – Dec 2021
Technologies: Arduino Uno, Embedded C, MQ-3 Sensor, IR Sensor, Relay, Buzzer

- Designed and developed a rider safety system by integrating helmet-worn detection, alcohol sensing, drowsiness detection, and alert mechanisms on an Arduino Uno, creating a comprehensive embedded safety prototype.
- Implemented the firmware architecture by programming sensor acquisition, threshold evaluation, relay control, and buzzer alerts in Embedded C, enabling automatic engine cutoff when unsafe conditions were detected.

Autonomous Robot, University of Sheffield Oct 2023
Technologies: Arduino, Embedded C, Sensors & Actuators, Fusion 360

- Built the complete electronic and mechanical integration for a 4-person autonomous robot by assembling the wiring, sensors, actuators, and chassis required for a multi-stage navigation challenge.
- Designed the robot structure in Fusion 360 by creating the chassis layout and mounting arrangements for the drive system, sensors, and robotic arm, producing a manufacturable and serviceable design.

EDUCATION

MSc Robotics, The University of Sheffield
Highlights: Redesigned an automated self-assembly platform using an 8-sided internal geometry, reducing assembly cycle time from 2–3 minutes to 1–1.5 minutes through controlled part rotation and extensive physical testing.

Bachelor of Technology in Mechatronics, Shri Vaishnav Institute of Technology and Science
Highlights: Developed the firmware and sensor-integration architecture for a Smart Helmet safety system, combining alcohol sensing, drowsiness detection, helmet-worn detection, automatic engine cutoff and rider alerts.

COURSES & CERTIFICATIONS

Industrial Automation Training – PLC Programming, HMI & SCADA Systems, TechMission 2022

- Completed hands-on training in Siemens S7 PLC programming, HMI development, and SCADA integration for industrial automation applications. Gained practical experience in panel wiring, electrical schematic interpretation, commissioning, and fault diagnosis of control systems.

Operations Industrial Engineer Job Simulation, Siemens 2026

- Completed a virtual simulation focused on industrial operations, process improvement, and manufacturing workflow optimisation. Applied structured problem-solving, operational analysis, and cross-functional decision-making to realistic engineering scenarios.